Functions of several variables.

[**SUBMIT ANSWERS TO ALL QUESTIONS**]

Total differential of $f(x,y,z)$: $df = \dfrac{\partial f}{\partial x}dx + \dfrac{\partial f}{\partial y}dy + \dfrac{\partial f}{\partial z}dz.$

Gradient vector: $\nabla f = \dfrac{\partial f}{\partial x}\boldsymbol{i} + \dfrac{\partial f}{\partial y}\boldsymbol{j} + \dfrac{\partial f}{\partial z}\boldsymbol{k} \equiv \left(\dfrac{\partial f}{\partial x}, \dfrac{\partial f}{\partial y}, \dfrac{\partial f}{\partial z}\right).$

Directional derivative along unit vector $\boldsymbol{T}$: $\dfrac{\partial f}{\partial s} = \nabla f \cdot \boldsymbol{T} = \dfrac{\partial f}{\partial x}T_x + \dfrac{\partial f}{\partial y}T_y + \dfrac{\partial f}{\partial z}T_z.$

Taylor's expansion for a function $f(x,y)$:

$$f(a+h, b+k) = f(a,b) + f'_x h + f'_y k + \frac{1}{2}\left[f''_{xx}h^2 + 2f''_{xy}hk + f''_{yy}k^2\right] + \dots,$$

where the derivatives $f'_x \equiv \partial f/\partial x$, $f'_y \equiv \partial f/\partial y$, $f''_{xx} \equiv \partial^2 f/\partial x^2$, etc., are evaluated at (a,b).

Stationary point: $f'_x(a,b) = 0$, $f'_y(a,b) = 0$. Let $A = f''_{xx}(a,b)$, $B = f''_{xy}(a,b)$, $C = f''_{yy}(a,b)$, and $\Delta = AC - B^2$. For $\Delta > 0$ this point is an extremum: $A > 0$ – minimum, $A < 0$ – maximum. For $\Delta < 0$ this is a saddle point. For $\Delta = 0$ further investigation is needed.

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1. Find the partial derivatives of the following functions:

$$\text{(a) } z = \frac{x-y}{x+y}, \quad \text{(b) } z = \sqrt{x^2 - y^2}, \quad \text{(c) } z = \ln\left(x + \sqrt{x^2+y^2}\right), \quad \text{(d) } z = \arctan\frac{y}{x}.$$

[20]

2. Find the gradient vectors of the following functions:

$$\text{(a) } f(x,y,z) = x^2 + y^2 + z^2 + xy + xz + yz, \quad \text{(b) } f(x,y,z) = \frac{1}{\sqrt{x^2+y^2+z^2}}.$$

[20]

3. Find the derivative of the function $f(x,y) = x^3 - 2x^2y + xy^2 + 1$ at point $P(1,2)$ in the direction from this point to point $Q(4,6)$. [20]

4. Find the second-order partial derivatives for $f(x,y,z) = \dfrac{1}{\sqrt{x^2+y^2+z^2}}$, and verify that

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}, \quad \frac{\partial^2 f}{\partial z \partial x} = \frac{\partial^2 f}{\partial x \partial z}, \quad \frac{\partial^2 f}{\partial z \partial y} = \frac{\partial^2 f}{\partial y \partial z}, \quad \text{and} \quad \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0.$$

[20]

5. Find the stationary points of the following functions and use the second-order derivatives to determine (if possible), whether they are maxima, minima, or saddle points:

$$\text{(a) } z = x^2 + xy + y^2 - 2x - y, \quad \text{(b) } z = e^{-(x^2+y^2)}, \quad \text{(c) } z = x^3 + y^3 - 3xy,$$

$$\text{(d) } z = 2x^4 + y^4 - x^2 - 2y^2.$$

[Hint: in (d) you need to find and investigate all nine(!) stationary points.] [20]